Canonical distinguishing colorings of cyclic tournaments

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Abstract

Albertson and Collins conjectured that every cyclic tournament of order $2q+1$ is distinguished by coloring $0, \dots, q$ with one color and the remaining vertices with a second color. We prove the stronger statement that, for every odd-order permutation group on $\mathbb{Z}_n$ containing the translation $x \mapsto x+1$, the set $\{0, \dots, (n-1)/2\}$ has trivial setwise stabilizer. The proof combines induction on block systems with the Burnside–Schur theorem. In the primitive case, intersection sizes of translates of the interval restrict an affine stabilizer to the identity or a reflection. The group theorem and its tournament corollary are also formally verified in Lean.

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1 Introduction

A coloring of a digraph is *distinguishing* if the identity is its only color-preserving automorphism. For an odd integer $n = 2q+1 \geq 3$, a *cyclic tournament* $T(n; S)$ has vertex set $\mathbb{Z}_n$ and arc $i \to j$ precisely when $j - i \in S$, where $S \subseteq \mathbb{Z}_n \setminus \{0\}$ contains exactly one member of each pair $\{d, -d\}$ of nonzero residues. Its automorphism group contains the translation $\tau : x \mapsto x+1$.

Albertson and Collins [1] proposed a prescribed distinguishing two-coloring for every cyclic tournament. In the formulation of Meslem and Sopena [5, Conjecture 2], the first color class is $I_n = \{0, \dots, q\}$. The original formulation [1, Conjecture 2.4] uses $\{0, \dots, q-1\}$; the statements are equivalent because the complement of this latter set is $I_n + q$, and translations are automorphisms. Meslem and Sopena proved the prescribed-coloring statement for several classes, including Paley tournaments, and whenever one of the two canonically induced subtournaments is rigid. As they explain, the existence of some distinguishing two-coloring already follows from Gluck's regular-subset theorem [4]. The issue here is the specified interval, independent of S.

A subset is *regular* for a permutation group when its setwise stabilizer is trivial. We prove the following group-theoretic statement.

Theorem 1.1. *Let n be a positive odd integer, and let $G \leq \mathrm{Sym}(\mathbb{Z}_n)$ have odd order and contain $\tau : x \mapsto x+1$. Then $I_n = \{0, 1, \dots, (n-1)/2\}$ is regular for G.*

2 The affine interval lemma

The following elementary observation treats all odd primes, including characteristic three, uniformly.

Lemma 2.1. *Let $p = 2q + 1$ be an odd prime and let $I = \{0, \dots, q\} \subseteq \mathbb{F}_p$. The setwise stabilizer of I in $\mathrm{AGL}_1(p)$ consists precisely of the identity and the reflection $x \mapsto q - x$.*

Proof. For $d \in \mathbb{F}_p$, put $C(d) = |I \cap (I + d)|$. Taking d to be its representative in $\{1, \dots, 2q\}$ gives

$$C(d) = q + 1 - \min\{d, p - d\}.$$

Indeed, for $1 \le d \le q$, the intersection is $\{d, \dots, q\}$; the remaining values follow from $C(-d) = C(d)$. In particular, among nonzero d the largest value is q, attained exactly at $d = 1, -1$.

Suppose $f(x) = \alpha x + \beta$, with $\alpha \neq 0$, preserves I. Since $f(I + d) = I + \alpha d$, bijectivity gives $C(\alpha d) = C(d)$. Taking $d = 1$ forces $\alpha = 1$ or -1. If $\alpha = 1$ and $\beta \neq 0$, then $I + \beta = I$ would give $C(\beta) = |I| = q + 1$, contrary to the displayed formula. Thus $\beta = 0$ in this case. The map $r(x) = q - x$ preserves I. If $\alpha = -1$, then $r \circ f$ is a translation preserving I, hence is the identity by the preceding case. Therefore $f = r$. □

3 Proof of the group theorem

Proof of Theorem 1.1. We use strong induction on odd n. The case $n = 1$ is immediate. Two observations will be used throughout: a subset and its complement have the same setwise stabilizer, and, since G contains every translation, regularity of I_n implies regularity of each translate $I_n + t$. For the latter assertion, conjugate a stabilizer by τ^t. The action of G is transitive because it contains τ.

Suppose first that G is imprimitive. Choose a nontrivial invariant block system and let B be the block containing 0. If $t \in B$, then $t \in B \cap (B + t)$, so the block property gives $B + t = B$. It follows that B is an additive subgroup of $\mathbb{Z}_n$. All blocks are translates of B. Hence for odd $a, b > 1$ with $n = ab$ the blocks are

$$B_r = \{r + at : 0 \le t < b\}, \qquad 0 \le r < a.$$

Write $a = 2u + 1$ and $b = 2v + 1$. Since $(n - 1)/2 = av + u$, we have

$$I_n \cap B_r = \begin{cases} \{r + at : 0 \le t \le v\}, & 0 \le r \le u, \\ \{r + at : 0 \le t < v\}, & u < r < a. \end{cases} \tag{1}$$

Let $g \in G$ preserve I_n. Its induced permutation on the blocks preserves their intersection sizes with I_n. The blocks with the larger size $v + 1$ are indexed precisely by I_a. The induced group on the blocks has odd order and contains $r \mapsto r + 1$ on $\mathbb{Z}_a$. The induction hypothesis therefore makes the induced permutation of g the identity. Thus g fixes every B_r setwise.

For each r, let H_r be the permutation group induced on B_r by its setwise stabilizer in G. It has odd order and contains the restriction of τ^a. In the coordinate $r + at \leftrightarrow t$ on $\mathbb{Z}_b$, this restriction is $t \mapsto t + 1$. If $r \le u$, the preserved subset in (1) is I_b. If $r > u$, its complement is $\{v, \dots, 2v\} = I_b + v$. In either case, induction, together with the complement and translation observations, makes the restriction of g to B_r the identity. Hence g is the identity.

It remains to treat the primitive case. The Burnside–Schur theorem in its permutation-action form [3, opening theorem] states that a primitive finite permutation group containing a regular cyclic subgroup is either 2-transitive or permutationally isomorphic to a subgroup of $\mathrm{AGL}_1(p)$ in prime degree p. A 2-transitive group of degree $n \ge 3$ has order divisible by $n(n - 1)$, which is even. Consequently $n = p$ is prime and the affine alternative holds.

We verify that the affine description applies in the original residue coordinates. Let $f : \mathbb{Z}_p \to \mathbb{F}_p$ be a bijection conjugating G into $\mathrm{AGL}_1(p)$. The permutation $f\tau f^{-1}$ has order p. An affine map with multiplier different from 1 has a fixed point and is conjugate to a multiplication, so its order divides $p-1$. Thus $f\tau f^{-1}$ is translation by some $c \neq 0$. It follows that $f(j) = f(0) + jc$ for every $j \in \mathbb{Z}_p$. The affine renormalization $z \mapsto (z - f(0))/c$ makes this bijection the original residue identification. Every element of G is therefore affine in those coordinates.

Lemma 2.1 now says that an element of G preserving I_p is either the identity or $x \mapsto q - x$. The latter is a nonidentity involution and cannot belong to an odd-order group. This completes the induction. □

4 Cyclic tournaments

Corollary 4.1. *For every cyclic tournament $T(2q+1; S)$ with $q \geq 1$, the coloring*

$$\lambda(i) = \begin{cases} 1, & 0 \leq i \leq q, \\ 2, & q < i \leq 2q \end{cases}$$

is distinguishing.

Proof. The group $\mathrm{Aut}(T)$ has odd order. Otherwise Cauchy's theorem would give a nonidentity involution, which swaps two vertices and therefore reverses the unique arc joining them, a contradiction. Moreover, $\mathrm{Aut}(T)$ contains τ. Theorem 1.1 implies that the only automorphism preserving the first color class I_{2q+1} is the identity, as required. □

5 Formal verification and reproducibility

Theorem 1.1 and Corollary 4.1 have been formally verified in Lean. The proof is available in the project repository at commit `2802921`. The declarations `canonicalHalfGroup_proved` and `tournament_half_regular`, in the namespace `AlbertsonCollins`, express the group theorem and the tournament conclusion, respectively. The latter assumes that rotation of the specified cyclic labels preserves all arcs; it concludes that the canonical half has trivial setwise stabilizer in the automorphism group.

The formal proof includes the block actions, strong induction, and the interval-intersection argument. The Burnside–Schur step uses proof terms from AxiomMath's formalization [2], with the source revision and Apache-2.0 license recorded in the repository. Thus the classification step is proved rather than assumed as an additional hypothesis. The complete project was built with Lean 4.35.0-rc3 and a pinned Mathlib revision. The final theorem axiom reports contain only `propext`, `Classical.choice`, and `Quot.sound`. The finite computational checks are not premises of either formal theorem.

Data availability. The results are proved analytically; no research datasets are used. The Lean proof, dependency pins, computational scripts, and graph examples are available in the project repository.

AI assistance. ChatGPT/Codex (OpenAI) assisted with checking the supplied proof, developing the interval-intersection argument, implementing finite checks, developing and debugging the Lean formalization, and drafting the exposition.

References

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